MECH 325 - Midterm Book 1
Closed Book

Thursday, October 26, 2023

Name: Solution

Student Number: _____

Circle your section

Section 102 (11:00- 12:30)

Section 101 (9:30 – 11:00)

Signature: _____

Part 1 MC Mark _____ / 12

Part 1 SA Mark _____ / 10

Part 2 LA Mark _____ / 38

Total _____ / 60

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (22 marks)

Multiple choice (12 marks: 1 marks per question). Complete all 12 questions by marking your response on the paper.

Short answer (12 marks). Complete all 5 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your paper in order to begin the Open Book portion. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (38 marks)

Long answer (38 marks): Complete all parts to the questions for Part 2 by marking your responses on the exam paper. This portion of the exam is open-book and open-notes. Laptops and tablets are permitted but must be disabled for communication and access to the internet.

NO COMMUNICATION DEVICES!

Part 1A: Multiple Choice (12 Points)

Answer all 12 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

1. **What is the best description of the type of motion between the teeth of two properly meshed bevel gears?**

   a) Purely rolling
   b) Purely sliding
   c) Some rolling when teeth engage and disengage, but predominantly sliding
   d) Some sliding when teeth engage and disengage, but predominantly rolling
   e) Approximately equal parts sliding and rolling

2. **For properly meshed worm gears which of the following statements is FALSE?**
   a) The magnitude of the axial force on the worm is the same as radial force on the gear
   b) The magnitude of the radial force on the worm is the same as the radial force on the gear
   c) The magnitude of the axial force on the worm is the same as the tangential force on the gear
   d) The magnitude of the tangential force on the worm is the same as the axial force on the gear
   e) The magnitude of the friction force on the worm is the same as the frictional force on the gear

3. **The worm gear shown has a 1.5" pitch diameter and has two threads. The gear has a 7" pitch diameter and 80 teeth. If the worm turns 20 revolutions in the clockwise direction indicated, how far does the gear turn?**

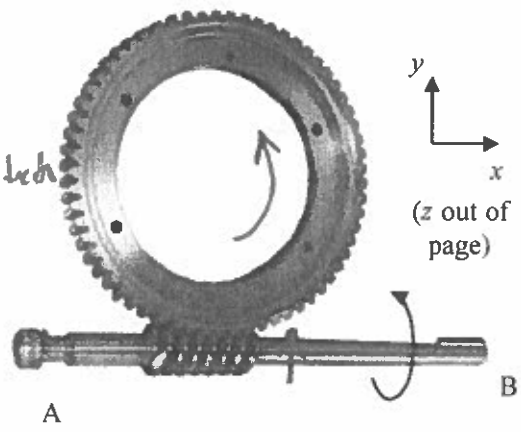
   a) ¼ rotation clockwise
   b) ¼ rotation clockwise
   c) ½ rotation clockwise
   d) ½ rotation counter-clockwise
   e) 4 rotations counter-clockwise

$N_w = 2$

$N_G = 80$

$20_{rev} = 20 \cdot 2 = 40\ teeth$

$\therefore \frac{1}{2}$ rev.



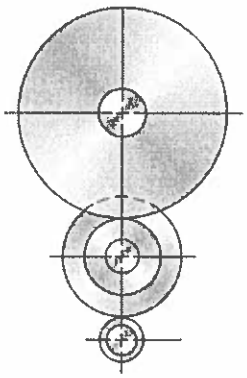
4. **Which of the following types of gears allows for the shafts to be offset and non-intersecting?**
   a) Bevel
   b) Spur
   c) Worm
   d) Hypoid
   e) Helical

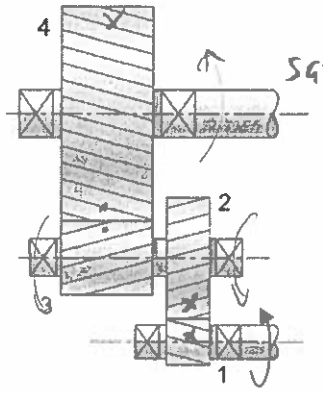
5. **A helical gear has a 20° pressure angle, a helix angle of 15°, a diametrical pitch of 8 and 20 teeth. What is the diameter of the gear (in inches)?**

$d = \frac{20 \text{ teeth}}{8 \text{ teeth/in}} = 2.5 \text{ in}$

a) 2.0
b) 1.5
c) 2.5
d) 3.0
e) 4.0

6. **For the gear train shown, Gear 1 is the driving pinion and has 15 teeth, Gear 2 has 32 teeth, Gear 3 has 24 teeth and Gear 4 has 75 teeth. An 1800 rpm motor is driving the gear 1 in the Clockwise direction (as shown). What is the speed and direction of the output shaft?**





same or 1

$e = \frac{15}{32} \cdot \frac{24}{75} = \frac{1}{4} \cdot \frac{3}{5}$

$e = 3/20$

a. 200 rpm clockwise
b. 200 rpm counter clockwise
c. 270 rpm clockwise
d. 270 rpm counter clockwise
e. 540 rpm clockwise

$n_{out} = e\, n_{in} = 3/20 \cdot 1800$

$= 270 \text{ rpm}$

7. **The motor attached to the gearset in Question 4 above is rated at 750 Watt. What is the torque on the output shaft?**

$P = T\omega \rightarrow T = P/\omega$

$T = 750 \text{ N-m/s} / (270 \text{ rpm} \times \frac{1}{60} \times (\frac{1}{\text{rad}}))$

$T = 26.5 \text{ N-m}$

a) 13.3 N-m
b) 19.9 N-m
c) 26.5 N-m
d) 35.7 N-m
e) 53.1 N-m

8. **The term *"Post-Extreme Horespower"* refers to the following:**

a) The breaking strength of a synchronous belt
b) The power rating for a chain where roller fatique or excessive wear are the modes of failure
c) The power rating for a chain where link fatigue failure is the primary mode of failure
d) The maximum horsepower rating for a chain
e) The smallest degree of lubrication

9. A V-belt drive has a driving pulley operating at 200 rpm, a drivetrain value of e = 0.25, a nominal power of H_{nom} = 1.0 hp, a service factor of K_s = 1.5, and a design factor of n_d = 1.25. TWO belts are to be used in parallel. The following belts are available in the catalogue. Which is the smallest belt that will satisfy the design requirements?

a) Allowable power H_a = 0.7 hp
b) Allowable power H_a = 1.1 hp
c) Allowable power H_a = 1.5 hp
d) Allowable power H_a = 2.9 hp
e) Allowable power H_a = 5.7 hp

$H_d = \frac{H_{nom} \cdot K_s \cdot n_d}{2}$
$= (1.0)(1.5)(1.25)/2$
$= 0.94 < 1.1$

10. Which of the following statements regarding gears is TRUE?

a) Spur gears can take higher loads than helical gears
b) Helical gears can never be used for right angle drives (shafts 90°)
c) Lower pressure angles allow for higher transverse forces between gears
d) Pinions with only 10 teeth can be used with a rack
e) Miter gears are a form of helical gears

11. Rank the following flexible drives from lowest to highest power capacity. Assume approximately equal dimensions and operating conditions.

a.	V-Belt	Flat belt	Roller Chain
b.	V-Belt	Roller Chain	Flat belt
c.	Flat belt	V-Belt	Roller Chain
d.	Flat belt	Roller Chain	V-Belt
e.	Roller Chain	Flat belt	V-Belt

12. When specifying the working load of wire rope, which of the statements is FALSE:

a. one must consider the modulus of elasticity of the wire rope
b. one must take in account the maximum allowable bearing pressure of the wire rope on the sheave
c. one must consider the application in order to specify a safety factor
d. one must determine the maximum sheave diameter
e. one must consider loads caused by sudden stops and starts

13 (Bonus) To convert from ft-lbs/min to horsepower use the following conversion

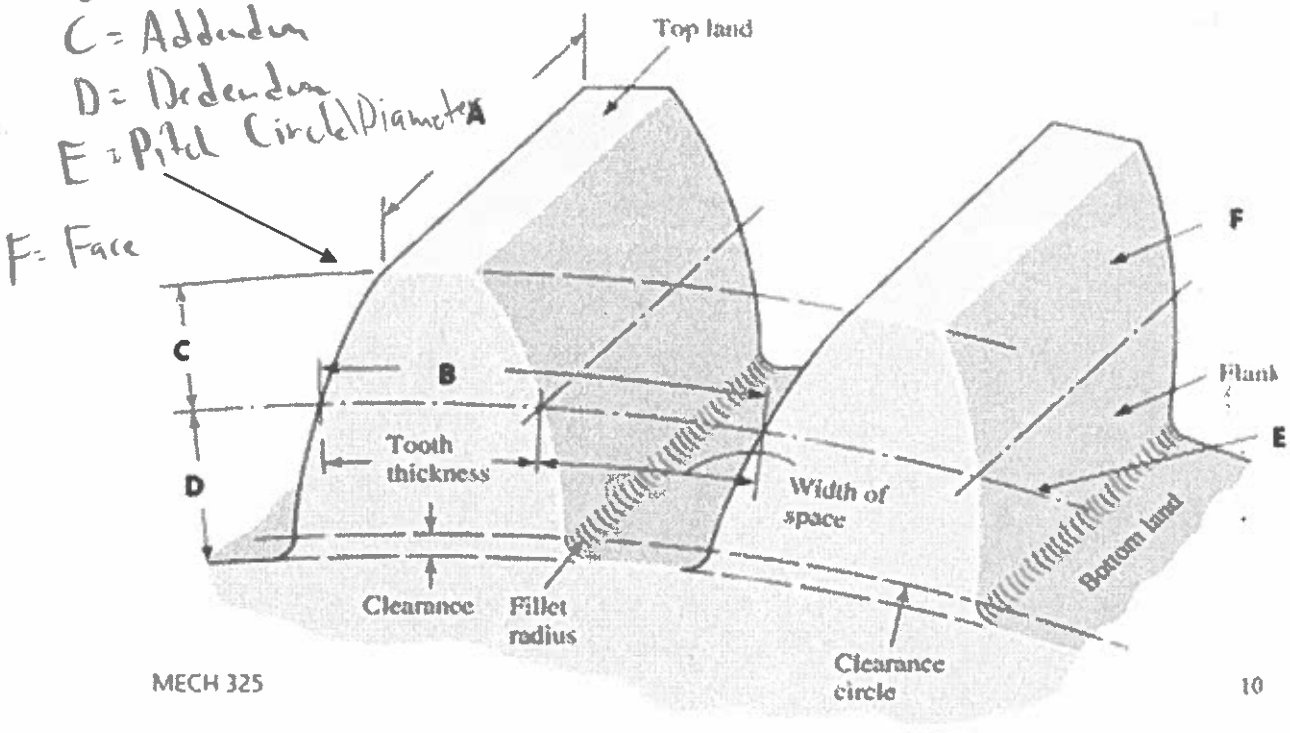
a) Multiply by 33000
b) Divide by 33000
c) Multiply by 550
d) Divide by 550
e) Divide by 454

Part 1B – Short-Answer Questions (10 marks)

Answer all questions in the spaces provided.

1. Name the SIX missing terms (A through F) related to spur gears (3 marks).

A = Face Width (will accept Crown or Depth)
B = Pitch
C = Addendum
D = Dedendum
E = Pitch Circle Diameter
F = Face



2. Which of the above terms are important for stress analysis of the gear teeth? Why ((3 marks)?

The pitch, pitch circle, and width are key terms for stress analysis.
Both contact & bending stresses ar width increases. A reduction in pitch increases tooth size which reduces bending stress. Increasing pitch circle will reduce transmitted forces (for same torque) which will reduce the stress.

3. What are the advantages of V-belts over chain drives? Are there any disadvantages? (2 marks)?

Advan: V-belts are friction drive and do not require lubrication
V-belts run quietly and can absorb a high degree of shock. They are higher in efficiency and do not exhibit cordal variation

Disadvan: V-belts require pre-tension in order to operate properly
V-belts come in only pre-determined lengths
V-belts do not work well for slow speed ($V < 1000$ fpm)
V-belts transmit less power than chains

4. What is meant by "through-hardening?" What is meant by "case-hardening?" Which one is preferred for highly loaded gears? Why (2 mark)

Through-hardening is when the entire component (gear) is heated above the critical temperature and then quenched in oil or water. The hard & brittle phase is then tempered to increase ductility. The entire component has the same mechanical properties (HB, S_y, S_{ult}, etc.)

Case-hardening focuses on only changing the properties of the surface of the component (gear, shaft etc.) This is done through concentrated heat (induction / flame) or increasing the local Carbon content through carburizing/nitriding.

Case hardening is best for highly loaded gears as these gears can withstand higher contact stresses.

MECH 325 – Midterm Book 2
Open Book

Thursday, October 26, 2023

Name: _____

Student Number: _____

Circle your section

Section 101 (11:00-12:30)

Section 102 (9:30-11:00)

I adhered to the policy of No Communication Devices.

Signature: _____

Instructions

Open notes and texts are allowed. Laptops are allowed but all communication features must be disabled.

No Communication Devices Allowed!

There are **Two** Problems to complete.

Part 2 – Open-Book (38 pts)

Long answer: Complete all parts to the question by marking your response in the pages provided. This portion of the exam is open-book and open-notes. **At the end of the exam, return this handout with your answers in the booklet.**

This is a multi-part problem. For each part, use the parameter values given. If you cannot complete one part, you may assume some values and move on in order to receive marks for subsequent parts. Clearly indicate your final answer to each part by drawing a box around it. Separate different parts with a line drawn across the page.

Long Answer Problem (38 marks)

Problem 1 (16 Marks)

A V-belt system is required to attach an electric motor to a conveyor system. The following information relates to the equipment

Driver	AC Motor (High Torque)
Driven Machine	Heavy Conveyor
Service	8 hours/day
Input Speed	1160 rpm
Input Power	50 HP
Output Speed (Nominal)	600 rpm
Minimum Centerline Distance	30 inches

Select a suitable V-belt belt system for the application. Specify the belt size, sheave sizes, and number of belts. What is the actual sheave speed and final center distance? What is the safety factor for this system?

Driver: AC Motor HT
Driven: Heavy Conveyor

Service Factor Moff Table 7-1 $K_s = 1.5$

(From Shigley Table 17-15 $1.4 < K_s < 1.6$)

2pt Design Power $H_D = K_s H_{nom} = 1.5(50\,HP) = 75\,HP$

From Figure 7-13 $N = 1160\,rpm$ & $H_D = 75HP$ we choose 5V or 5VX belt

Velocity Ratio $= VR = \frac{1160}{600} = 1.933$ close to 2.0

1pt Determine sheave sizes for velocity ratio using Fig. 7-15

Let's limit belt speed to $V = 5000\,ft/min$.

$V = \frac{\pi D n}{12} = 5000 = \frac{\pi D (1160)}{12}$ gives $D_1 < 16.4''$

Options include:

D_1	D_2	VR
13.9	27.7	1.99
13.1	27.7	2.11
12.4	21.1	1.70
11.7	21.1	1.80
10.8	21.1	1.95
10.2	21.1	2.06

2pt

← Select $D_1 = 13.9''$, $D_2 = 27.7$ as best option

[Note: other combinations are acceptable]

Selection of $D_1 = 13.9''$ gives $H_{tab} \doteq 30HP$ [Figure 7-15] @ 1160 rpm

(1 pt) Multiple belts will be required.

Output speed $= 1160/1.99 = \boxed{583 \text{ rpm} = N_{output}}$

Specify a trial center distance CD Min CD = 30" (given)

$D_2 < CD < 3(D_2 + D_1)$ Eq'n 7-17

$27.7'' < CD < 3(27.7 + 13.9) \Rightarrow 27.7 < CD < 124.8''$

(2 pts) Let's choose $CD = 32''$ for this application

Perimeter Length $L_p = 2CD + 1.57(D_2 + D_1) + \frac{(D_2 - D_1)^2}{4CD}$ Eq'n 7-12

$L_p = 2(32) + 1.57(27.7 + 13.9) + (27.7 - 13.9)^2/4(32)$

$= 64.0 + 65.312 + 5.200 = 134.5$ inches

From Table 7-2 for 5V belts one can choose 132" or 140"

Select 132" for this application

Actual $CD = \frac{B + \sqrt{B^2 - 32(D_2 - D_1)^2}}{16}$ $B = 4L_p - 6.28(D_2 + D_1)$ Eq'n 7-13

(1 pt) $B = 4(132) - 6.28(27.7 + 13.9) = 266.75$

$CD = \left(266.75 + \left[(266.75)^2 - 32(27.7 - 13.9)^2\right]^{1/2}\right)/16$

$\boxed{CD = 32.61''}$ Final Center Distance

Wrap Angle $\theta_1 = 180° - 2\sin^{-1}\left(\frac{D_2 - D_1}{2CD}\right)$ Eq'n 7-14

(1 pt) $= 180 - 2\sin^{-1}\left(\frac{27.7 - 13.9}{2(32.61)}\right)$

$\boxed{\theta_1 = 155.6°}$ s. $\theta_2 = 204.4°$

$\boxed{\text{Belt Length } L_p = 132 \text{ inches}}$

(1 pt) Corrected Power Rating Per belt = $H_{allowable} = H_a$

$$H_a = C_\theta C_L (H_{tab} + \Delta P)$$

From Fig. 7-17 $\Delta P = 1.26$ @ 1160 rpm for $V = 2.0$

(3 pt) From Fig. 7-18 $C_\theta \doteq 0.94$ @ $\theta_1 \simeq 155°$

From Fig 7-19 $C_L \doteq 1.02$ @ $L_p = 132''$

From Fig 7-15 $H_{tab} = 30$ HP @ $D_1 = 13.9''$ & $N_1 = 1160$ rpm

$$H_a = (0.94)(1.02)(30 + 1.26) = \underline{29.97 \text{ HP} = H_{allowable}}$$

Minimum Number of Belt: $N_B \times H_a > H_d$

$N_B \times 29.97 > 75$

(2 pt) Min belts $N_B = 3.0$

Safety Factor $n_{FS} = \frac{H_{all}}{H_d} = \frac{(3.0)(29.97)}{75} = 1.20$

Select: Three 5V belts Length = 132"

$D_1 = 13.9''$ $D_2 = 27.7''$

Conveyor Speed = 593 rpm

CD = 32.61"

Safety factor = 1.20

Problem 2 (22 Points)

Part 1 of 3

(8 points) A straight bevel gear pair is required to drive a reciprocating pump (moderate shock). The pinion has $N_P = 20$ teeth, $P_d = 10$ teeth/inch, 20° pressure angle, and face width, $F = 1.00$ inch. The pinion is driven at 1750rpm by an AC motor (normal torque) delivering 10HP. The recommended life for the machine is 1500 hours of operation. Assume the pinion is outboard mounted (cantilevered) and the gear is straddle mounted and both are of commercial gear quality ($A_v = 8$, $Q_v = 9$).

Design a set of bevel gears for this application.

Note: you can choose imperial or metric units

Driver	AC Motor (Normal Torque)
Driven Machine	Pump (Reciprocating)
Service Life	1500 hours
Input Speed	1750 rpm
Output Speed	580 rpm (nominal)
Input Power	10 HP (7.5 kW)
Pressure Angle (Normal)	20°
Diametrical Pitch	10 teeth/inch (module = 2.5 mm.)
Face Width	1.00 inch (25mm.)
Pinion	20 teeth
Gear	? teeth

Determine the following:

a) Given 20 teeth for the pinion, select a gear and determine the gear ratio (velocity ratio)
b) The pitch diameters and associated cone angles
c) Pitch line speed
d) Determine the tangential, radial, and axial forces for the gears. Determine these based on the pitch diameters and not the average (mean) diameter. You will need these for part 2.

(2pt) a) $N_p = 20$ $N_G = ?$ Gear Ratio $= 1750\ rpm / 580\ rpm = 3.02 \sim 3.0$

$\therefore m_G = 3 = N_G / N_p$ $\boxed{N_G = 60 \text{ teeth for } m_G = 3}$ $N_G = 61$ teeth permitted

b) Pitch $= 10$ teeth/inch $D_p = 20 \text{ teeth} / P_d = 20/10 =$ $\boxed{D_p = 2.0 \text{ in}}$

$D_G = 60/10$ $\boxed{D_G = 6.0 \text{ in}}$

(2pt) Cone angle $\gamma = \tan^{-1}(N_p/N_G)$ [Table 8-8]

$= \tan^{-1}(20/60)$ $\gamma = \boxed{18.4° = \gamma}$

$\Gamma = 90 - \gamma = \boxed{71.6° = \Gamma}$

(1pt) c) Pitch Line Speed $V_t = \frac{\pi D_p n_p}{12} = \frac{\pi(2)(1750)}{12}$

$V_t = 916.3$ ft/min

Note $V_t = 916.3$ fpm < 6867 fpm max for $A_v = 8$ gears OK ✓

d) Forces on bevel gears

(3pt) $W_{tp} = T/(d/2)$ with $d = D_p = 2$ inches

and $T = 63000\ P/n$

$W_{tp} = 63000\ P/n / d/2 = 63000(10HP)/1750rpm/2''/2$

$W_{t_p} = 360$ lbf

$W_{Pr} = W_{tp} \tan\phi \cos\gamma = 360 \tan 20° \cos 18.4° = 124.3$ lbf

$W_{Paxial} = W_{tp} \tan\phi \sin\gamma = 360 \tan 20° \sin 18.4° = 41.4$ lbf

tangential force Pinion = $W_{tp} = W_{tG} = 360$ lbf

axial force pinion = $W_{xp} = W_{rG} = 41.4$ lbf

radial force pinion = $W_{rp} = W_{tG} = 124.3$ lbf

Part 2 of 3

(9 Points) Determine the AGMA **bending** stress number for the 20 tooth **pinion ONLY.**

Given:

Driver	AC Motor (Normal Torque)
Driven Machine	Pump (Reciprocating)
Service Life	1500 hours
Input Speed	1750 rpm
Output Speed	580 rpm (nominal)
Input Power	10 HP (7.5 kW)
Pressure Angle (Normal)	20°
Diametrical Pitch	10 teeth/inch (module = 2.5 mm.)
Face Width	1.00 inch (25mm.)
Gear Quality	$A_v = 8$ ($Q = 9$)
Material for Pinion & Gear	Steel
Pinion Mounting	Outboard (Cantilevered)
Gear Mounting	Straddle (bearings on either side of gear)
Overload Factor	**$K_o = 1.50$** (uniform power source and moderate shock)
Pinion bending stress number	s_t = ?

Use ACMA procedure for bending stress Mott 10-9
Shigley Fig 14-7

(1 pt)

$$\sigma = s_t = \frac{W_t P_d}{F J} K_o K_s K_m K_v$$

From Part 1) $W_t = 360$ lbs

Pitch $P_d = 10$ teeth/inch

Face width $F = 1.0$ inch

(2 pt)

From Fig 10-15 (Shigley 15-15) 20 teeth meshing with 60 teeth
$\phi = 20°$ shaft angle 90°

$N_p = 20$ $N_G = 60$ $J = 0.25$

$K_o = 1.5$ given

(1 pt) [Fig 10-13] Size Factor $K_s = 0.4867 + 0.2132/P_d = 0.508 = K_s$

(1 pt) [Fig 10-14] Load Dist. Factor $K_m = K_{mb} + 0.0036F^2$

$K_{mb} = 1.10$ for one gear straddle mounted

$K_m = 1.10 + 0.0036[1^2] = 1.104 = K_m$

Vel. Factor $K_v = ?$

Fig 9-16 $V_t = 916.3$ ft/min [Part 1] & $A_v = 8$

$K_v = 1.18$ from Figure

or $B = 0.25(A_v - 5)^{0.667} = 0.25(8-5)^{0.667} = 0.520$

$C = 50 + 56(1.0 - B) = 50 + 56(1 - 0.52) = 76.88$

(1 pt)

$$K_v = \left[\frac{C}{C + \sqrt{V_t}}\right]^{-B} = \left[\frac{76.88}{76.88 + \sqrt{916.3}}\right]^{-0.52} = 1.189$$

$K_v = 1.19$

$\therefore$ $$S_t = \frac{(360)(10)}{(1)(0.25)}(1.5)(0.508)(1.104)(1.189)$$

(2 pt)

$S_t = 14,396$ psi $= 14.4$ ksi

Part 3 of 3

(5 points) Select a suitable through hardened steel for the **pinion**. Specify the alloy and heat treatment.

The gear is expected to operate for 1500 hours. Define your expected reliability and temperature factors.

What is your safety factor for the pinion bending stress? Assume the same material will be selected for the gear.

Comment on the selection of the proposed bevel gears. Is the analysis complete?

Required permissible bending stress S_{at}

$$S_{at} = \frac{S_t (SF)(K_R)(K_T)}{K_L}$$ Shigley

(1pt) assume reliability is 99%, so $K_R = 1.0$

assume $T < 120°(250°F)$ so $K_T = 1.0$

assume safety factor $SF = 1.0$

Stress cycle K_L

(1pt) Number of load cycles $N_L = (1500\ hr)(1750\ rev/min)(60\ min/hr)$

$N_L = 1.576 \times 10^8$

Fig 10-12 $K_L = 1.3558 N_L^{-0.0178}$ $K_L = 0.969$

$$S_{at} = \frac{(14396\ psi)(1)(1)(1)}{(0.969)} = 14857\ psi = 14.9\ ksi$$

Fig 10-17 Grade 1 Gear

Required $HB = (S_{at} - 2.10)/0.044 = (14.857 - 2.10)/0.044$

Required $HB = 290$

(1pt) Table 9-8 Typical alloys for through hardening 1045, 4140, 4150, 4340, 4350

Let's select AISI 4140 OQT 1000 $HB = 341$

$\therefore S_{at} = 0.044 HB + 2.10 = 0.044(341) + 2.10 = 17.1\ ksi$

(1pt) $SF = \frac{S_{at} K_L}{S_t K_R K_T} = \frac{17.1(0.969)}{14.4(1)(1)}$

$SF = 1.15$

Comment: With a S.F of 1.15 the pinion gear is suitable for application based ONLY on bending stress.

However analysis is not complete. One must determine the contact stress number. Our experience has shown that the failure is most likely due to contact stresses and not bending stress.